$$\lim_{x\to c} f(x) = \lim_{x\to c}\sqrt{1-x^2} = \left[\lim_{x\to c}(1-x^2)\right]^{\frac{1}{2}}$$

$$= (1-c^2)^{\frac{1}{2}} = f(c)\,.$$

$$\lim_{x\to -1^+} f(x) = \lim_{x\to -1^+}(1-x^2)^{\frac{1}{2}} = 0 = f(1)\,.$$

$$\lim_{x\to -1^-} f(x) = \left[\lim_{x\to -1^-}(1-x^2)\right]^{\frac{1}{2}} = 0 = f(-1).$$

Thus f is continuous on $[-1, 1]$. One can also solve this problem using composite function theorem.

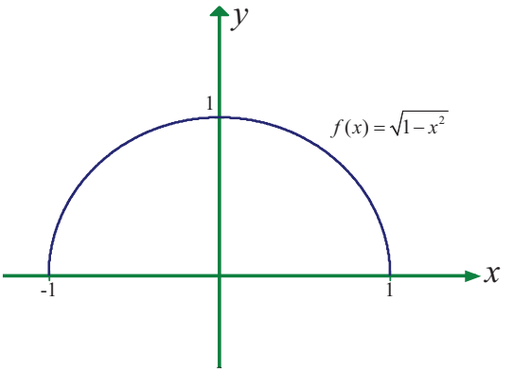


Fig. 9.36

Example 9.36

Describe the interval(s) on which each function is continuous.

(i) $f(x) = \tan x$

(ii) $g(x) = \begin{cases} \sin\dfrac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$

(iii) $h(x) = \begin{cases} x\sin\dfrac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$

Solution

(i) The tangent function $f(x) = \tan x$ is undefined at $x = (2n+1)\dfrac{\pi}{2}, n \in Z.$

At all other points it is continuous, so $f(x) = \tan x$ is continuous on each of the open intervals

$$\ldots\left(-\frac{3\pi}{2}, -\frac{\pi}{2}\right), \left(-\frac{\pi}{2}, \frac{\pi}{2}\right), \left(\frac{\pi}{2}, \frac{3\pi}{2}\right), \ldots$$

(ii) The function $y = \dfrac{1}{x}$ is continuous at all points of $\mathbb{R}$ except at $x = 0$ where it is undefined.

The function $g(x) = \sin\dfrac{1}{x}$ is continuous at all points except $x = 0$, where $\lim\limits_{x\to 0} g(x)$ does not exist. So, g is continuous on the intervals $(-\infty, 0)$ and $(0, \infty)$

(iii) The function $h(x)$ is defined at all points of the real line $\mathbb{R} = (-\infty, \infty)$; for any $x_0 \neq 0$,

$$\lim_{x\to x_0} h(x) = \left(\lim_{x\to x_0} x\sin\frac{1}{x}\right)$$

$$= x_0 \sin\frac{1}{x_0} = h(x_0)$$

For $x_0 = 0$

$$h(x) = x.\sin\frac{1}{x}$$

$$-x \leq x\sin\frac{1}{x} \leq x$$

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